Christina M. Amend

Bryan, TX • (682) 360-1125 • christinaamend@tamu.edu

EDUCATION

Bachelor of Science, Geographic Information Sciences & Technology **Expected: Dec 2025**
Minor in Ecology & Conservation Biology
Texas A&M University, College Station, TX

RELEVANT WORK & PROJECT EXPERIENCE

Site Suitability Analysis Using Terrain and Hydrological Modeling January 2025
Advanced GIS Course Project
Performed site suitability analysis in ArcGIS Pro by deriving DEM-based terrain layers (slope, aspect, hillshade), modeling hydrological patterns (flow direction, accumulation, watershed), and integrating factors through weighted overlay and raster algebra to identify optimal locations

Spatial Patterns of Sea Turtles in the Gulf of Mexico January 2025
Data Analysis in Geography Course Project
Conducted kernel density estimation (KDE) and hotspot mapping in ArcGIS Pro, processed shapefiles and visualized data in R programming, and applied generalized additive models (GAMs) with Kruskal–Wallis testing to analyze sea turtle spatial distributions

Compilation of Environmental, Threat, and Animal Data for Cetacean Population Health Analysis Platform (CETACEAN) November 2024
GCOOS Project
Developed a WebGIS platform in ArcGIS Online integrating machine learning workflows, spatial data infrastructure, and custom Python tools to process, centralize, and visualize high-volume multi-source marine mammal datasets for research and restoration in response to the Deepwater Horizon Oil Spill

Geometric Rectification / Orthorectification January 2024
Remote Sensing in Geosciences Course Project
Rectified raw digital images using image-to-map rectification, image-to-image rectification and GPS/GNSS ground control rectification methods to generate orthoimagery in ENVI using aerial photography and satellite imagery

Student Research Assistant **November 2024 - Current**
Gulf of America Coastal Ocean Observing System, Texas A&M University
- Developed CETACEAN Data Portal, a WebGIS platform integrating high-volume marine mammal datasets with custom Python tools for centralized analysis and visualization
- Worked on a team to build the Sea Turtle Atlas, a centralized WebGIS platform consolidating multi-agency datasets on distributions, habitats, and threats to support large-scale restoration planning and conservation management
- Developed GIS infrastructure and web-based data tools for Gulf of America Coastal Acidification Network (GCAN) to integrate ocean acidification monitoring datasets, improve data accessibility, and support to collaborative research and management

Student Research Assistant **June 2024 - November 2024**
Biological and Agricultural Engineering Department, Texas A&M University
- Evaluated and validated Esri deep learning models for geospatial analysis in cotton agriculture, assessing model performance and applicability to spatial datasets
- Developed custom web mapping applications for senior capstone courses using Esri Experience Builder, leveraging widgets, APIs, and data integration workflows
- Instructed graduate students in foundational GIS concepts, including spatial data management, geoprocessing, and cartographic visualization

ACTIVITIES & ACHIEVEMENTS

- Texas A&M Chapter of Citizens Climate Lobby, Bryan/College Station (BCS) Chapter of CCL, Aug 2024 -
- Geography Society, Fall 2023 -
- Dean's Honor Roll Spring 2024, Distinguished Student Spring 2024

SKILLS

- Esri software, ENVI (Digital Image Processing), GitHub, MS Office suite
- Familiar with programming languages Python, R, C++, Java, SQL

PORTFOLIO

cmaPortfolio